Project Overview
This project focuses on the design and implementation of a two-wheeled self-balancing ultrasonic obstacle-avoidance transport vehicle based on an Arduino microcontroller. The work combines embedded system development, sensor integration, control system design, and system testing.

Hardware Design
The system is plan to built around the **Arduino Uno**. It gets 8-bit 16 MHz processor and only 2 KB of SRAM, may struggle to handle simultaneous tasks such as balance control, encoder feedback, IMU sampling, and Bluetooth communication. As an alternative choice, the **Arduino Nano ESP32** offers higher clock speed and larger memory, making it more suitable for real-time multitasking control and providing better balance performance. This project will prioritize the use of **Arduino Uno**. If its performance is insufficient, alternative solutions will be adopted.

Except for the main control unit, modular components are adopted:
The **TB6612FNG** dual motor driver controls two **JGA25-370 DC** motors with encoders, which provide movement and speed feedback.
A **MPU6050** inertial measurement unit (IMU) supplies balance and motion data, while dual incremental encoders record wheel rotation.
An **HC-SR04** ultrasonic sensor detects obstacles and measures distance in front of the vehicle.
Wireless control is achieved through an **HC-05** Bluetooth module, which communicates with an Android smartphone application. The system is powered by a 12 V lithium battery, with an **LM2596** buck converter providing a regulated 5 V supply for the microcontroller, where the 12 V line powers the motors, and the 5 V line powers the microcontroller.

Software and Control
All subsystems are programmed in **Arduino C++**. Each functional block—motor control, encoder speed measurement, Bluetooth communication, ultrasonic sensing, and IMU data acquisition—has been developed and tested individually. The next stage is to integrate these modules into a unified control framework. A PID control algorithm will be implemented to maintain balance, with parameters tuned experimentally for stability and responsiveness. Additional logic for turning and navigation will allow the vehicle to move forward, reverse, and avoid obstacles.
Additional software requires: **Android Studio** for making an android bluetooth communication apps. **PCB software** is used to draw schematic and PCB layout. **CAD software** to model the car framework.

Sensor Fusion and Testing & Safety Considerations
Data from the MPU6050 (angle and angular velocity) and encoders (wheel speed) will be fused to provide accurate feedback for real-time control. Testing will be performed on both flat and uneven surfaces to assess balance performance, obstacle-avoidance accuracy, and response time.

All hardware components operate within safe voltage and current limits. No hazardous materials are involved. The primary safety concern is electrical isolation between the motor and logic circuits, which ensures stable operation and protects both the hardware and the user.